where it can be used again.

When we apply force to the brake pedal, the car slows down and the motor works in reverse direction. When running in invalidate direction, the motor acts as the generator and, thus, charges the battery. Using regenerative braking in an EV reduces cost of fuel, increasing fuel financial system and lowering emission. The regenerative braking system provides braking force while speed of the vehicle is low and, hence, the traffic stop-and-go. Thus, deceleration required is less in an EV.

Drive system. The drive system transfers mechanical energy to the traction wheels in generating motion. It makes the transmission system unnecessary in an electrical vehicle.

Charging station. EVs are available in a variety of models with different ranges and capabilities. To recharge, these are plugged into a source of electric power through electric vehicle supply equipment (EVSE). EV charging stations—also called electric recharging points, charging points, charge points and EVSE—are an element in an infrastructure that supplies electric energy for the recharging of EVs, such as plug-in EVs. These stations are also needed when travelling, and many support faster charging at higher voltages and currents than are available from residential EVSE.

Types of charging

The various types of charging are described below.

Level 1 charging. Level 1 charging uses the same 120V current found in standard household outlets in the USA and can be performed using the power cord and equipment that most EVs come with. Low installation cost is its advantages, while slow charging (typically 4.8 to 8.04 kms, or 3 to 5 miles, of range per hour) is the disadvantage.

Level 2 charging. Level 2 charging uses 240V power to enable faster regeneration of an EV's battery system. This type of charging requires installation of an EVSE unit and electrical wiring capable of handling higher voltage power. It has a faster charge time (16.09 to 32.2 kms, or 10 to 20 miles, of range per hour). It is more energy-efficient than level 1 charging, but is a little more expensive.

DC fast charging (480V). DC fast charging provides compatible vehicles with an 80 per cent charge in 20 to 30 minutes by converting high-voltage AC power to DC power for direct storage in EV batteries. Automakers currently have three specifications for DC fast charging plugs, CHAdeMO (or CHArge de Move), SAE Combined Charging System (CCS) and Tesla Supercharger standards.

Nissan and Mitsubishi vehicles use CHAdeMO, while many current and upcoming vehicles from American and European manufacturers have SAE CCS ports. Tesla's Supercharger equipment is only compatible with Tesla Model S or later vehicles, although the company is developing an adaptor that will allow Tesla owners to use CHAdeMO equipment.

Charge time has reduced drastically—it is nearly as fast as refuelling a petrol vehicle. It is significantly more expensive than level 1 or 2 equipment, and high-voltage 3-phase power connections to utilities further increase installation costs. Also, it has potential issues with cold weather operation.

EV charging in India

The Indian government has finally announced a policy for the rollout of EV charging infrastructure. In an official document that was sent out by Ministry of Power on December 14, 2018, the government has outlined several key facts, figures and a plan to support the expansion of EVs in the country. Push for electric mobility charging stations will first be rolled out in cities with a population of greater than four million residents, that is Mumbai, New Delhi, Bengaluru, Hyderabad, Ahmedabad, Chennai, Kolkata, Surat and Pune.

The government has also announced that in the cities mentioned above, there will be at least one EV charging station in a grid of 3km. On the highway, there will be an electric car station every 25km. The government has also recognised key corridors that will have electric charging stations for both small private vehicles and large commercial vehicles.

The most important announcement in the circular is that the setting up of these charging stations will be a de-licensed activity and, thus, rollout is expected to be quite rapid. Charging stations will also be free to obtain electricity from any power company through the open access system.

While there are no restrictions on private charging stations being set up in a house/apartment complex/office block, public charging stations have to meet minimum requirements. Every charging station will have to have a minimum of three fast chargers, one CCS-style plug, one CHAdeMO (equivalent to move using charge) and one type-2 AC fast charger. While the first two plugs will have to give a minimum output of 50kW and 200V to 1000V, the type-2 plug will have to be a minimum of 22kW and 380V to 480V.

In addition, charging stations will also compulsorily have two slow/moderate charge points, one with Bharat DC-001 connection with 15kW and 72V to 22V, and the other, Bharat AC-001 with 10kW and 230V.

To propel the growth of EVs in India, the government recently slashed the applicable rate of Goods and Services Tax (GST) on lithium-ion batteries. Many states have also geared up for the uptake of EVs as a means of mainstream mobility. EFY

Batteries used in Tesla

The battery that Tesla has been using, sourced from Panasonic, for its Model S electric car is most likely lithium-ion with a cathode that is a combination of lithium, nickel, cobalt and aluminium oxide. The battery industry calls this an NCA battery, and it has been around—and made by Panasonic—for many years now.

Typically, lithium-ion NCA batteries use a combination of 80 per cent nickel, 15 per cent cobalt and five per cent aluminium. It is unclear what mix Tesla and Panasonic's battery combination is. (Anodes in these traditional lithium-ion batteries are usually a graphite combination, which acts as a host for lithium-ions.) Addition of aluminium to NCA batteries makes them more stable.